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- Show Name: Intro to IoT Pentesting
- Topic Name: IoT Basics
- Episode Name: Hardware Components

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Hardware Components

Objectives:

- Identify hardware components commonly found in IoT devices

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- UART
 - Universal Asynchronous Receiver-Transmitter
 - Serial Protocol (like RS232)
 - Devs use it for debugging
 - Can be used to get root shell
 - Usually 4 hardware pins/pads
 - Can be more, but only 3 are needed
 - GND
 - RX
 - TX
 - Other pins
 - VCC
 - UART Frame
 - Idle high (1)
 - Start bit low (0)
 - Data bits 5-9 bits
 - Parity bit error checking (rarely used)
 - Stop high (1)
 - 8N1
 - 8 Data Bits
 - No Parity
 - 1 Stop bit
 - JTAG
 - Joint Test Action Group
 - Allows dev to debug hardware on multi-layered circuit boards
 - Usually 10 Pins, but 5 are needed
 - TCK Test Clock Input
 - TMS Test Mode Select
 - TDI Test Data Input
 - TDO Test Data Output
 - TRST Test Reset Input
 - Others
 - VCC
 - GND

- It is typical to see multiple GND pins
 - This helps to mitigate crosstalk on a flat ribbon cable
- SOIC
 - Small Outline Integrated Circuit
 - 4, 8, 16, 24 pins
 - SPI
 - Serial Peripheral Interface
 - Can connect to retrieve data from EEPROM/Flash Memory
 - Uses 4 connections/wires to transmit data
 - CS Chip Select
 - SCK Serial Clock
 - MOSI Master OUT, Slave IN
 - MISO Master IN, Slave OUT